English Pronunciation Improvement Assistant

Interactive Python application · speech practice, recognition, comparison and audio feedback

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THE PRACTICE GAP

Pronunciation practice needs immediate feedback

English learners often struggle with pronunciation and speaking confidence.

Traditional practice does not always show what went wrong right away.

This project turns a speaking exercise into an interactive feedback loop.



The assistant turns speech into actionable feedback

Speak

The student reads a reference word into a microphone.

Recognize

The application converts the recording into text.



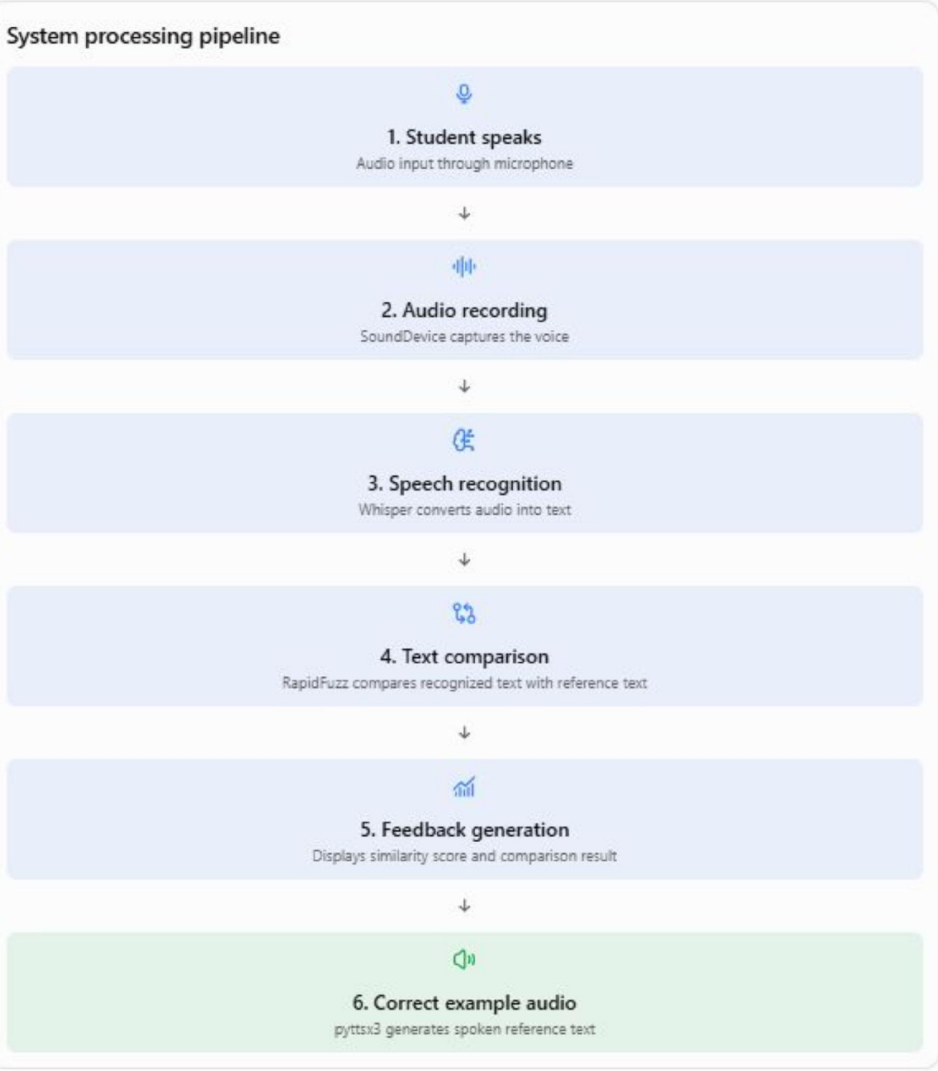
Compare

The recognized text is checked against the reference.

Improve

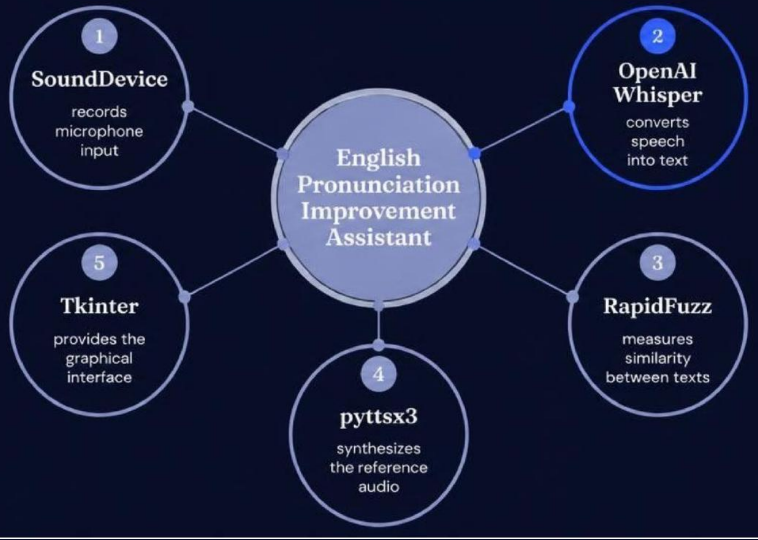
The student listens to an example and tries again.

Proposed processing pipeline



IMPLEMENTATION STACK

Five technologies work together in one pipeline



Text matching is a useful start—not a complete pronunciation assessment

Current limitation: speech recognition can produce incorrect transcriptions.

Current limitation: text similarity does not directly measure phonetic pronunciation accuracy.

Future direction: add phoneme-level analysis, word-stress detection, and more advanced pronunciation assessment.

Thank you